

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Explain principle of construction, operation and characteristics of a MOSFET with applications

Q.24 Explain the working principle of

- a) Nickel-Cadmium batteries.
- b) Silver-oxide batteries

Q.25 What is the effect of alternating voltage applied to pure resistance, pure inductance and pure capacitance.

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Roll No.

1st Year / Computer Engg.

Subject : Fundamentals of Electrical & Electronics Engineering

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 FET stands for

- a) Farad effect Transistor
- b) Field Effect Transistor
- c) Field Emitter Transistor
- d) none of these

Q.2 The winding of a transformer made up of

- a) Copper
- b) Aluminium
- c) Nickel
- d) Both a and b

Q.3 In lead acid battery, negative plate is made up of

- a) PbSO_4
- b) Pb
- c) PbO
- d) PbO_2

Q.4 The frequency of AC is

- a) 0 Hz b) 50 Hz
- c) 100 Hz d) 120 Hz

Q.5 The unit of Capacitance is

- a) Henry b) Ohm
- c) Farad d) none of these

Q.6 In N-type semiconductor which type of impurity is added

- a) Pentavalent b) Trivalent
- c) None of these d) a and b both

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 CMOS stands for _____

Q.8 What do you mean by transistor biasing?

Q.9 What is admittance?

Q.10 JFET stands for _____

Q.11 Define Inductance.

Q.12 Draw the symbol of NPN Transistor.

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Explain conductance, susceptance and impedance with their units.

Q.14 Define power factor and its significance.

Q.15 How electro-magnetic field is produced by flow of electric current?

Q.16 What are the principles of self & mutual induction?

Q.17 Difference between AC and DC?

Q.18 What is series and parallel resonance and its condition?

Q.19 Explain the concept of current growth, decay and time constant in an RL circuit.

Q.20 Explain Kirchoff's Voltage Law.

Q.21 What are the care and maintenance of lead-acid battery?

Q.22 What is the working principle of Lead-Acid batteries?